MILITARY SPECIFICATION

MODULE ASSEMBLIES, FIRING AND SAFING, TRAINING

1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of firing or safing module assembly used with the Atomic Demolition Charge Trainer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Method of Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-S-60902 - Soldering, Electrical Connections for Nuclear Weapons Systems, Procedure for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inpsection by Attributes
MIL-STD-810 - Environmental Test Method
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipments of Inert Ammunition Components
MS35206-210- Screw

FSC 1125

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DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8823355 - Connector Receptacle Electrical
8823360 - Mount, Fuse
8823588 - Plate
8823589 - Training, Firing Squib Switch Module Assembly (Wired)
8823590 - Training Safing Switch Module Assembly (Wired)
8823602 - Rubber, Silicone
8823603 - Curing Agent
8823683 - Rubber, Silicone RTV-60
8823687 - Training, Firing Squib Switch Module Assembly (Unwired)
8823688 - Training, Safing Squib Switch Assembly (Unwired)
8823696 - Fuse
8824974 - Box, Set-up, Paperboard, for Squib activated switch, Training
8858963 - Carton, Packing, Ammunition, for Switch, Squib Activated; Fire or Switch Squib Activated, Training: Fire
8883534 - Bag packing for Repair Part
8886889 - Sleeve
9209786 - Primer
9209787 - Sealant
9209789 - Silicone Fluid
ET-8823589 - Equipment Tabulation
ET-8823590 - Equipment Tabulation
ET-8823687 - Equipment Tabulation
ET-8823688 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The module assembly shall comply with all requirements specified on the drawings (dwg) 8823589 or 8823590 or 8823687 or 8823688, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Circuit resistance. The circuit resistance shall comply with table I when tested as specified in 4.4.1.

Table I

Circuit resistance (see dwg 8823589)

From Pin	To Pin	Ohms Max
10	18	0.2

Table II

Circuit resistance (see dwg 8823590)

From Pin	To Pin	Ohms Max
1	2	0.2
11	22	0.2

3.6 Insulation resistance. The insulation resistance of any circuit shall not be less than 20 megohms when subjected to a potential of 500 ± 50 volts direct current (vdc) with the current limited to 5 milliamperes (ma) when tested as specified in 4.4.2. After the humidity test, the value of resistance shall be 1.0 megohm minimum.

3.7 Environmental testing. The module assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The module assembly shall also comply with 3.5 and 3.6 as applicable before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

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3.7.1 Low temperature. The module assembly shall withstand exposure to a storage temperature of -65 ± 5 degrees fahrenheit (°F) for a duration of 48 hours minimum and shall comply with 3.5 and 3.6 as applicable while at -65 ± 5°F when tested as specified in 4.4.3.1.

3.7.2 Humidity. The module assembly shall withstand exposure to a humid atmosphere when tested as specified in 4.4.3.2.

3.8 Marking adhesion. The marking adhesion of the assembly shall comply with Paint adhesion, section 3, TT-C-490 when tested as specified in 4.4.4.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.4). When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwgs 8823589 or 8823590 or 8823687 or 8823688) and 2 sets of parts for first article testing to the testing facility designated in the invitation for bids or request for proposal.

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4.2.2 Inspection and testing to be performed. The first article insepction sample shall be subject to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

Table III

First Article Inspection and Tests

Inspection or test	Contractor's Inspection Permitted Before Submission	Dest	Non Dest	Parts	Assembly
1. Classification of Characteristics (see 4.3.2.1)	X		X	X	X
2. Circuit resistance (see 3.5)	X		X		X
3. Insulation resistance (see 3.6)	X		X		X
4. Low Temperature (see 3.7.1)	X		X		X
5. Humidity (see 3.7.2)		X			X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Connector, receptacle, electrical (see dwg 8823355, rubber silicon (see dwg 8823602), curing agent (see dwg 8823603), rubber silicon RTV-60 (see dwg 8823683), fuse (see dwg 8823696), primer (see dwg 9209786), sealant (see dwg 9209787), silicone fluid (see dwg 9209789), screw (see MS35206-210), covering details of dwgs 8823589, 8823590, 8823687, and 8823688.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence (see 6.2) for component in accordance with applicable drawing or standard.....	Visual
Minor:	None defined.	

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4.3.2.1.2 Sleeve (see dwg 8886889) covering a detail of dwgs 8823589, 8823590, 8823687, and 8823688 as applicable.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length	Gage
	102. Diameter	Gage
	103. Thickness	Gage
	104. Material (see 3.1)	Visual
Minor:		
	201. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.3 Plate (see dwg 8823588) covering a detail of dwgs 8823589, 8823590, 8823687, and 8823688.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of tapped holes, max.	Gage
	102. Minor diameter of tapped holes, max.	Gage
	103. Material (see 3.1)	Visual
Minor:		
	201. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.4 Mount, fuse (see dwg 8823360) covering a detail of dwgs 8823589 and 8823687.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence (see 6.2) for fuse mount in accordance with dwg 8823360	Visual
	102. Length of fuse mount base	Gage
	103. Marking	Visual
	104. Material (see 3.1)	Visual
Minor:		
	201. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.5 Training firing squib switch module assembly (unwired) (see dwg 8823687) and training firing squib switch module assembly (wired) (see dwg 8823589).

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Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length	Measure*
	102. Width	Measure*
	103. Location of fuse mount	Gage
	104. Height of fuse mount	Gage
	105. Depth of connector, max	Gage
	106. Location of connector	Gage
	107. Location of sleeves	Gage
	108. Soldering in accordance with MIL-S-60902(MU) is applicable before potting	Visual/Manual
	109. Potting in accordance with dwg 8823687 or 8823589	Visual
	110. Marking	Visual
	111. Depth of chamfer	Gage
	112. No connector pins damaged	Visual
	113. Sleeves flush or below flush	Visual/Manual

*Measure with suitable scale.

Minor:		
	201. Serialization (see 3.4)	Visual
	202. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.6 Training safing switch module assembly (wired) (see dwg 8823590) and safing switch module assembly (unwired) (see dwg 8823688).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of connector	Gage
	102. Location of sleeves	Gage
	103. Length	Measure*
	104. Width	Measure*
	105. Depth of connector, max	Gage
	106. Soldering in accordance with MIL-S-60902(MU) if applicable	Visual/Manual
	107. Depth of chamfer	Gage
	108. No connector pins damaged	Visual
	109. Potting in accordance with dwg. 8823590 or 8823688	Visual
	110. Sleeves flush or below flush	Visual/Manual

*Measure with suitable scale.

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Categories	Characteristics	Method of Inspection
Minor:		
	201. Serialization (see 3.4)	Visual
	202. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.7 Box, set-up, paperboard, for squib activated switch, training (see dwg 8824974 and Section 5.3).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Contents tight in box	Manual
	102. Corners blunted	Visual/Manual
	103. Marking	Visual
	104. Air exhausted from bag	Manual
	105. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
	106. Packaging materials (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.8 Carton packing, ammunition, for switch, squib actuated: fire or switch, squib actuated, training: fire (see dwg 8858963 and Section 5.2).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Spacer placed over clips	Visual
	102. Contents tight in box	Manual
	103. Corners blunted	Visual/Manual
	104. Marking	Visual
	105. Air exhausted from bag	Manual
	106. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
	107. Packaging materials (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.9 Bag, packing, for repair parts (see dwg 8883534 covering dwg part number 8883534-9 and Section 5.2).

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Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Material (see 3.1)	Visual
	102. Length before packing, min	Measure*
	103. Width before packing, min	Measure*
	104. Objective evidence (see 6.2) available for heat seal test in accordance with MIL-P-116	Visual
	105. Properly packed	Visual
	106. No tear, split, or crack in bag	Visual

*Measure with suitable scale.

Minor:		
	201. Marking	Visual
	202. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.10 Packing and marking for storage and shipments (see dwg 8808362 and Section 5.2).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Gross load weight, max	Measure*
	102. Contents tight in package	Manual
	103. Correctly packed	Visual/Manual
	104. Package material (see 3.1)	Visual

*Determine weight with a suitable scale.

Minor:		
	201. Marking	Visual
	202. Workmanship (see 3.9)	Visual/Manual

4.3.2.1.11 Preparation for delivery for production parts (see Section 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Categories	Characteristics	Method of Inspection
Major:		
	101. Preservation, packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual/Manual
Minor: None defined.		

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

	Test	Test Classification
a.	Circuit resistance	Major
b.	Insulation resistance	Major
c.	Marking adhesion	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance (see 3.5), insulation resistance (see 3.6) and marking adhesion (see 3.8). Each assembly shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.4 as applicable. If any item fails to comply with 3.5, 3.6, and 3.8, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8823589 or ET-8823590 or ET-8823687 or ET-8823688. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. An deviation from equipment or procedures specified on the EL as desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (see 3.5). The circuit resistance shall be measured in accordance with Tables I and II as applicable (see 3.5) with approved measuring equipment.

4.4.2 Insulation resistance (see 3.6). The insulation resistance shall be measured with approved insulation resistance measuring equipment in accordance with Tables VI and VII, as applicable.

Table VI

Insulation resistance (see dwg 8823589)

Positive Potential to	Negative Potential to all other pins, shell, sleeves and fuse clips except
Pin 10	Pin 18

Table VII

Insulation resistance (see dwg 8823590)

Positive Potential to	Negative Potential to all other pins, shell and sleeves except
Pin 1	Pin 2
Pin 11	Pin 22

4.4.3 Environmental testing (see 3.7).

4.4.3.1 Low temperature (see 3.7.1). The module assembly shall be tested in accordance with MIL-STD-810, Method 502.

4.4.3.2 Humidity (see 3.7.2). The module assembly shall be tested in accordance with MIL-STD-810, Method 507, Procedure I.

4.4.4 Marking adhesion (see 3.8). The marking adhesion shall be tested in accordance with paint adhesion, section 4, TT-C-490.

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5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The module assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The module assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The module assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A.

5.2.1.1.1 Module assemblies (see dwg 8823687 or 8823589). The module assembly shall be packaged in accordance with dwg 8838963 and 8808362 using the specified moisture barrier.

5.2.1.1.2 Module assemblies (see dwg 8823590 or 8823688). The module assembly shall be packaged in accordance with dwgs 8883534 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C.

5.2.1.2.1 Module assemblies (see dwg 8823687 or 8823589). The module assembly shall be packaged in accordance with dwg 8838963 and 8808362 except that the moisture barrier may be omitted.

5.2.1.2.2 Module assemblies (see dwg 8823590 or 8823688). The module assembly shall be packaged in accordance with dwg 8883534 and 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C.

5.2.2.1.1 Module assemblies (see dwg 8823687 or 8823589). Module assembly shall be packed in accordance with dwg 8838963 and 8808362.

5.2.2.1.2 Module assemblies (see dwg 8823590 or 8823688). Module assembly shall be packaged in accordance with dwgs 8883534 and 8808362 as applicable.

5.2.3 Marking.

5.2.3.1 Module assemblies (see dwg 8823687 or 8823589). Module assembly shall be marked in accordance with 8838963 and 8808362.

5.2.3.2 Module assemblies (see dwg 8823590 or 8823688). Module assembly shall be marked in accordance with 8883534 and 8808362 as applicable.

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5.3 End item XM95 firing device.

5.3.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.3.1.1 Level A (see dwgs 8823687 and 8823688). The module assemblies shall be packaged in accordance with dwg 8824974 and 8808362 using the specified moisture barrier.

5.3.1.2 Level C (see dwgs 8823687 and 8823688). The module assemblies shall be packaged in accordance with dwg 8824974 and 8808362 except that the moisture barrier may be omitted.

5.3.2 Packing.

5.3.2.1 Level A and C (see dwgs 8823687 and 8823688). The module assemblies shall be packed in accordance with dwg 8824974 and 8808362.

5.3.3 Marking (see dwg 8823687 and 8823688). Marking shall be in accordance with dwg 8824974 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Production or repair part or end item XM47 Firing Device.
c. Selection of applicable level of packaging and packing required (see 5.1, 5.2, and 5.3).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1667 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-A.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table III) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table III) are verified by a representative designated by the Government.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2350 (Rev 1), dated 18 Aug 1963.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1125-A055

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